

Agenda

1. Announcements

- HW 12.1, 12.2 due today
- Discussion Quiz 10 tomorrow
- Midterm 2 Regrade Requests due today
- Check for final exam conflicts

2. Section 12.4

Recall:

$$\vec{v} = \langle x_1, y_1, z_1 \rangle \quad \vec{w} = \langle x_2, y_2, z_2 \rangle$$

$$\vec{v} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = |y_1 z_2| \vec{i} - |x_1 z_2| \vec{j} + |x_1 y_2| \vec{k}$$

$$= (x_1 z_2 - y_2 z_2) \vec{i} - (x_1 z_2 - x_2 z_1) \vec{j} + (x_1 z_2 - x_2 y_1) \vec{k}$$

Example: Let $\vec{v} = \langle 2, 1, -1 \rangle$ and $\vec{w} = \langle -3, 4, 1 \rangle$. Find $\vec{v} \times \vec{w}$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -1 \\ -3 & 4 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 4 & 1 \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & -1 \\ -3 & 1 \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & 1 \\ -3 & 4 \end{vmatrix} \vec{k}$$

$$= (1 - (-4)) \vec{i} - (2 - 3) \vec{j} + (8 - (-3)) \vec{k}$$
$$= 5 \vec{i} + \vec{j} + 11 \vec{k} = \langle 5, 1, 11 \rangle$$

Activity: Let $\vec{v} = \langle 2, 1, -1 \rangle$ and $\vec{w} = \langle -3, 4, 1 \rangle$. Find $\vec{w} \times \vec{v}$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 4 & 1 \\ 2 & 1 & -1 \end{vmatrix} = (-4 - 1)\vec{i} - (-3(-1) - 2)\vec{j} + (-3(1) - 2(4))\vec{k} \\ = \langle -5, -1, -11 \rangle$$

Observe: $\vec{w} \times \vec{v} = -(\vec{v} \times \vec{w})$

The cross product is orthogonal to both vectors.

Properties of the Cross Product

Theorem $\vec{v} \times \vec{w}$ is orthogonal to both $\vec{v}$ and $\vec{w}$.

$$\vec{v} \cdot (\vec{v} \times \vec{w}) = 0 \quad \vec{w} \cdot (\vec{v} \times \vec{w}) = 0$$

Theorem: If θ is the angle between two vectors $\vec{v}$ and $\vec{w}$, then $|\vec{v} \times \vec{w}| = |\vec{v}||\vec{w}|\sin\theta$

Stick with $\vec{v} \cdot \vec{w} = |\vec{v}||\vec{w}|\cos\theta$ to find θ
(Easier computations)

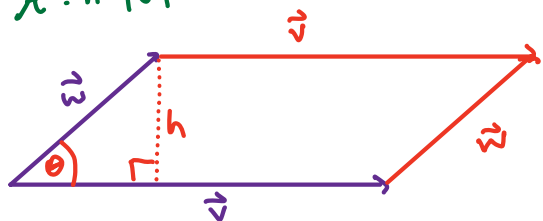
Properties Let $\vec{v}$, $\vec{u}$, and $\vec{w}$ be vectors in $\mathbb{R}^3$. Need $\mathbb{R}^3$ for $\times$ product
Let c be a constant.

- 1.) $\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$ 2.) $\vec{v} \times (\vec{u} + \vec{w}) = \vec{v} \times \vec{u} + \vec{v} \times \vec{w}$
3.) $(\vec{v} + \vec{w}) \times \vec{u} = \vec{v} \times \vec{u} + \vec{w} \times \vec{u}$ 4.) $c(\vec{v} \times \vec{w}) = (c\vec{v}) \times \vec{w} = \vec{v} \times (c\vec{w})$
5.) $\vec{v} \cdot (\vec{w} \times \vec{u}) = (\vec{v} \times \vec{w}) \cdot \vec{u}$ 6.) $\vec{v} \times (\vec{w} \times \vec{u}) = (\vec{v} \cdot \vec{u})\vec{w} - (\vec{v} \cdot \vec{w})\vec{u}$
7.) $\vec{v} \times \vec{v} = \vec{0}$

Theorem: Two vectors $\vec{v}$ and $\vec{w}$ are parallel if and only if $\vec{v} \times \vec{w} = \vec{0}$.

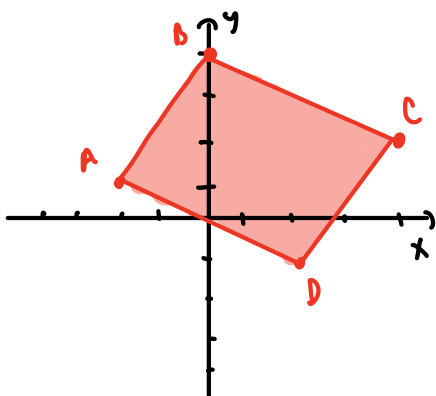
Area of Parallelograms

$$A = h \cdot |\vec{v}|$$



$$\sin \theta = \frac{h}{|\vec{w}|} \quad h = |\vec{w}| \sin \theta \quad \underline{A = |\vec{v}| |\vec{w}| \sin \theta = |\vec{v} \times \vec{w}|}$$

Example: Find the area of the parallelogram with vertices $A(-2, 1)$, $B(0, 4)$, $C(4, 2)$, and $D(2, -1)$.



in $\mathbb{R}^3$

$$\begin{aligned} A &= (-2, 1, 0) \\ B &= (0, 4, 0) \\ C &= (4, 2, 0) \\ D &= (2, -1, 0) \end{aligned}$$

$$\vec{AB} = \langle 2, 3, 0 \rangle$$

$$\vec{AD} = \langle 4, -2, 0 \rangle$$

$$\text{Area} = |\vec{AB} \times \vec{AD}|$$

$$= |\langle 0, 0, -16 \rangle| = 16$$

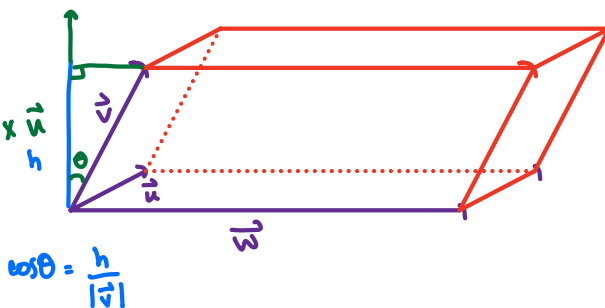
Scalar Triple Product

Three vectors $\vec{v}$, $\vec{w}$, and $\vec{u}$ in $\mathbb{R}^3$ that are not coplanar span a solid called a parallelepiped.

$$\text{Volume} = |\vec{w} \times \vec{u}|$$

$$= \underbrace{|\vec{w} \times \vec{u}|}_{\text{area of base}} \underbrace{|\vec{v}| \cos \theta}_{\text{height}}$$

$$= |\vec{v} \cdot (\vec{w} \times \vec{u})|$$



Activity: Find the volume of the parallelepiped spanned by the vectors $\langle 1, 0, 0 \rangle$, $\langle 1, 1, 0 \rangle$, and $\langle 1, 0, 1 \rangle$.